

Contents

1 Chapter 02 — Game Theory & CFR Basics: Implementation Report	1
1.1 1. What was developed	1
1.2 2. Key Results and Visualizations	1

1 Chapter 02 — Game Theory & CFR Basics: Implementation Report

Environment: April 2026

Game: Kuhn Poker (3-card, 2-player)

Algorithm: Counterfactual Regret Minimization (CFR) with chance sampling

Targets: Nash strategies to 4 decimal places, game value $\approx -1/18$, exploitability $O(1/\sqrt{T})$

Status: Game-value and convergence-rate targets met; mixed frequencies within 0.005–0.038 of the closed form (4-decimal accuracy reached only for the King’s and the Jack’s pure decisions)

1.1 1. What was developed

Chapter 02 implements a modular version of Counterfactual Regret Minimization (CFR) with chance sampling, applied to Kuhn Poker, written from scratch in Python. It proves that the framework successfully discovers Nash equilibrium strategies in a multi-agent, imperfect-information environment.

Key Components Built: - **Game Engine:** `kuhn_poker.py` handles terminal payoffs, betting actions, and info-set representations. - **CFR Core Algorithm:** Recursive tree traversal with Chance Sampling and Regret Matching (`cfr_trainer.py`, `info_set_node.py`). - **Exploitability & Best Response Evaluator:** Brute-force pure strategy aggregation to calculate exact exploitability (`evaluate/best_response.py`, `evaluate/exploitability.py`). - **Convergence Logger & Visualizations:** Utilities to plot exploitability and strategy convergence metrics over iterations (`utils/plotting.py`, `evaluate/convergence.py`). - **OpenSpiel Cross-Verification:** Check against standard open-source libraries (`compare_openspiel.py`).

1.2 2. Key Results and Visualizations

- **Minimax Optimal Convergence:** The algorithm converged to within 0.038 of the analytical Nash equilibrium family (parameterized by α); the exploitability of the average strategy after 100,000 iterations is 0.006.
- **Exploitability Decay Rate:** The log-log slope for exploitability over iterations is -0.52 ± 0.02 (5 seeds, 100–100,000 iterations), consistent with the theoretical $O(1/\sqrt{T})$ bound.

- **Game Value Accuracy:** After 100,000 iterations, the exact value of the average strategy (-0.0555) matches the theoretical value $-1/18$ (-0.0556) to within $8 \cdot 10^{-5}$; the running mean of the sampled payoffs is -0.0613 .

1.2.1 Game Value Convergence

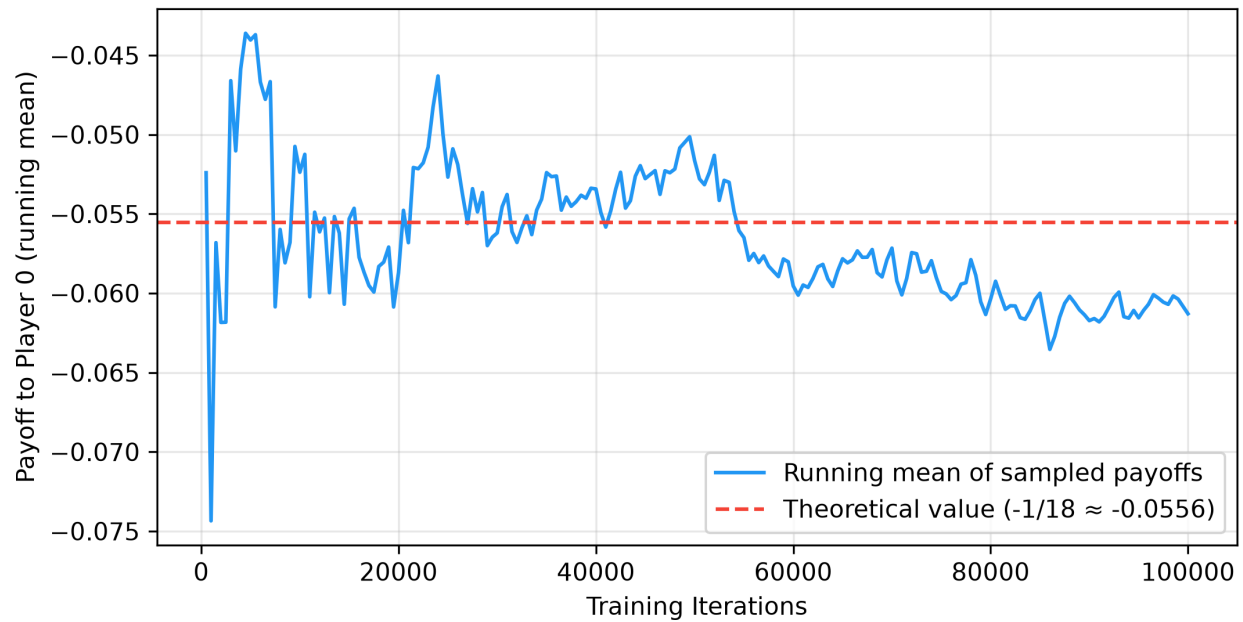


Figure 1: Game Value Convergence

1.2.2 Exploitability Convergence

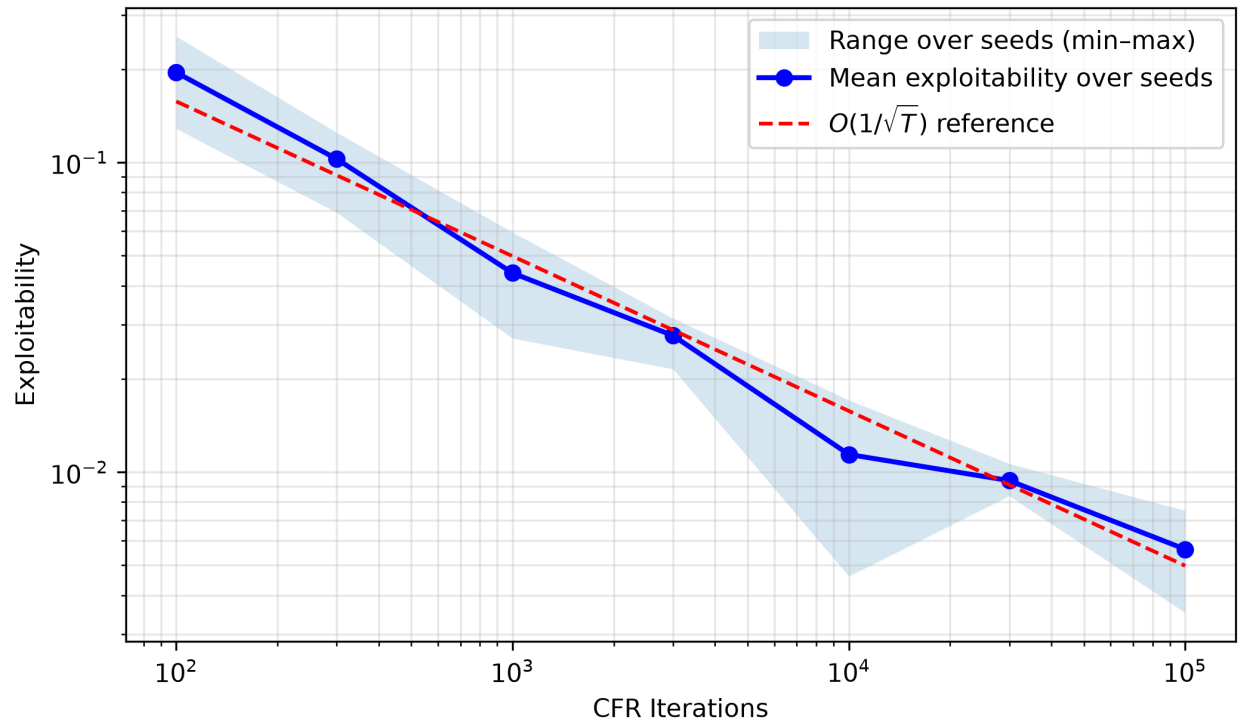


Figure 2: Exploitability Convergence

1.2.3 Strategy Analysis

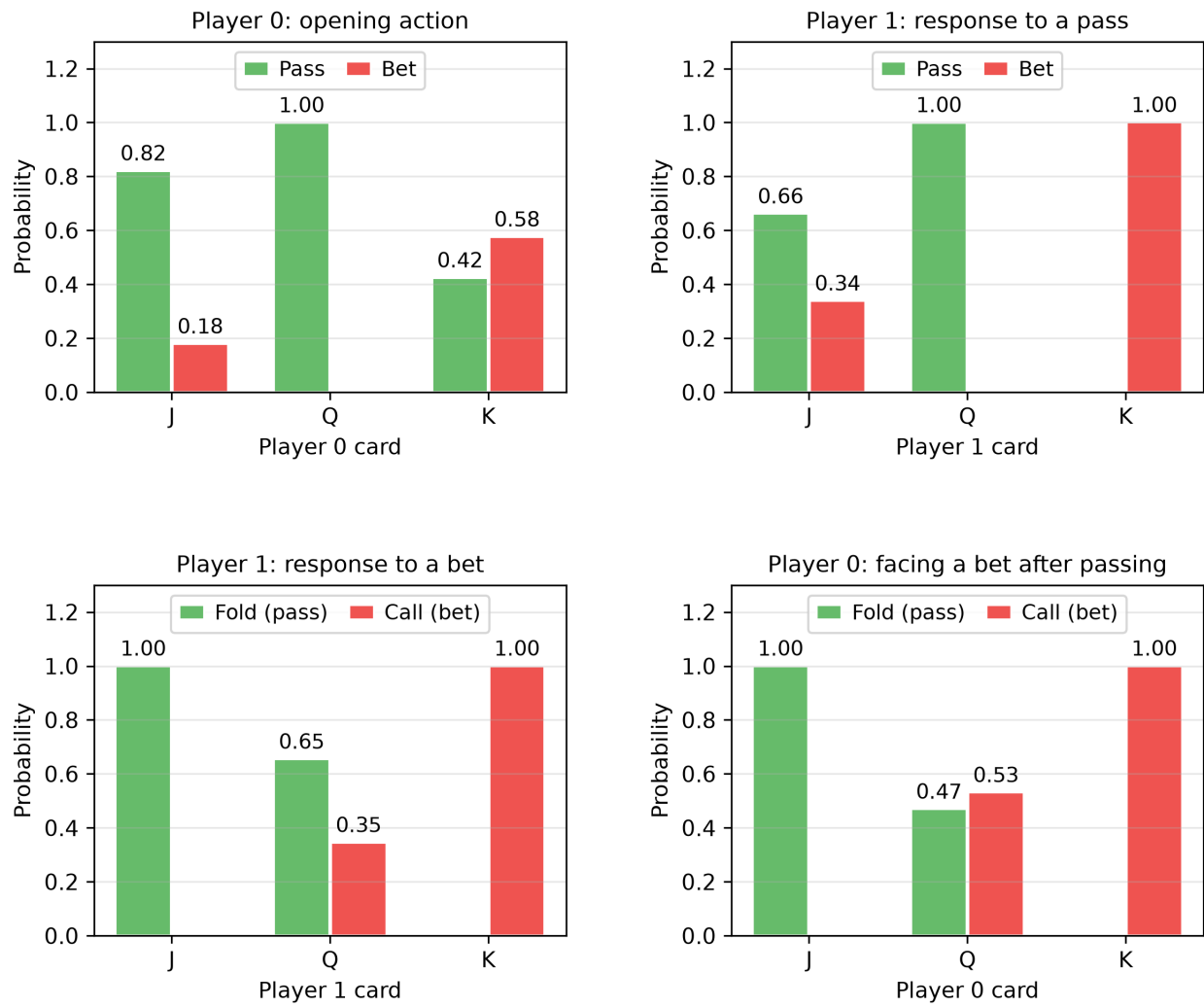


Figure 3: Strategy Analysis